Mazen Emam

Frontend Engineer | React.js · TypeScript | M.Sc. Student at FAU
Buckenhof, Germany | German: A0 (Beginner)
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LinkedIn | GitHub | Medium | Portfolio

PROFILE

Frontend engineer with 4 years shipping production React/TypeScript for manufacturing, threat-analytics, and recruitment platforms across MENA and Europe - including a production AI chat feature querying 300+ live sources, plus AI/LLM engineering across two self-directed projects. M.Sc. student in International Information Systems at FAU Erlangen-Nürnberg, authorized to work in Germany. Targeting Frontend/Software Engineer roles, part-time or full-time.

EDUCATION

Friedrich-Alexander-Universität (FAU) · Erlangen - Nürnberg, Germany ***Oct 2026 – Present***
M.Sc. International Information Systems

Information Technology Institute (ITI) *Sep 2020 – Dec 2020*
MEARN Track, 3-Month Diploma — bridge into software engineering

Alexandria University · Alexandria, Egypt *Oct 2013 – Jul 2017*
B.Sc. Management Information Systems

WORK EXPERIENCE

Siren Analytics · Beirut, Lebanon *Jun 2025 – Jan 2026*
Software Engineer | React.js · TypeScript · Material UI · React Query · Redux Toolkit · WebSocket · SockJS/STOMP · React Hook Form · Yup

- Led development of an AI chat module enabling users to query 300+ news sources conversationally, replacing manual content navigation across bulk-ingested datasets.
- Implemented role-based access control across a multi-tenant architecture, scoping feature and content visibility per tenant and enforcing user-level permissions platform-wide.
- Engineered a multi-language translation system with React Query caching and bulk translation hooks, supporting the company's expansion into the European market.
- Pioneered a real-time WebSocket notification system using SockJS/STOMP with throttled error handling, enabling instant cross-user threat flagging that navigates recipients directly to flagged content.
- Optimized performance by integrating React Compiler and leading application-wide code splitting, reducing unnecessary re-renders and improving bundle efficiency.
- Secured authentication by refactoring JWT expiry logic and eliminating race conditions in token refresh, preventing unauthorized access.
- Eliminated a class of runtime errors by establishing a TypeScript enforcement pipeline with strict ESLint rules and a pre-push hook, catching type inconsistencies before code integration.

Indos - Garment IO · Giza, Egypt *May 2023 – Jun 2025*
Mid-Senior Frontend Engineer | React.js · TypeScript · Material UI · Redux · Sentry · Formik · Yup · AWS · GitHub Actions

- Introduced TypeScript and strict ESLint enforcement across a 6-year-old codebase spanning 10+ modules and 40+ pages via an incremental migration strategy, establishing type safety without disrupting active development.
- Modernized the build pipeline by migrating from Create React App to Vite, reducing production build time by 50% and improving local development speed by 5x.
- Designed an i18n system with real-time language switching across 4 languages, enabling product deployment across MENA and European markets.
- Implemented Sentry performance monitoring capturing Web Vitals and API latency bottlenecks, identifying production issues across 30+ manufacturing facilities.
- Built a QR code scanning system for line managers, embedding machine IDs into scannable codes and eliminating manual lookup across 30+ facilities.
- Resolved performance bottlenecks and memory leaks by converting sequential API calls to parallel execution, removing unused data fetching, and enforcing pagination on unbounded endpoints.

AsgaTech · Giza, Egypt *Nov 2022 – May 2023*

Frontend Engineer | React.js · TypeScript · Material UI · MobX · React Hook Form · Yup

- Established a reusable component library with RTL support for the company's recruitment platform, creating a consistent design system that reduced new feature development time by 50%.
- Accelerated app startup speed by 35% through image optimization, lazy loading, and efficient API integrations.

Akwad · Alexandria, Egypt *Oct 2021 – Jul 2022*

Frontend Engineer | React.js · JavaScript · Bootstrap · Context API · React Hook Form

- Created a reusable form and grid component framework, standardizing development and reducing page build time by 90%.
- Overhauled the core data grid infrastructure, unifying pagination and search logic to eliminate inconsistent behavior and reduce bugs.

PROFESSIONAL DEVELOPMENT & RELOCATION

Independent Projects & Relocation *Feb 2026 – Oct 2026*

- Continued developing and maintaining independent software projects, including Job Radar and Echo FM, while strengthening my frontend and full-stack development skills.
- Secured admission to the M.Sc. International Information Systems program at FAU; secured visa and finalized relocation logistics ahead of program start.

PROJECTS

Job Radar — AI-Powered Job Dashboard

GitHub – Live | *Next.js 14 · TypeScript · React · Tailwind CSS · Supabase (PostgreSQL) · Google Gemini API · GitHub Actions*

- Built the entire user-facing product in Next.js 14 and TypeScript, including a personalized job dashboard, a multi-step settings and onboarding flow, an application tracker, and a salary insights explorer.
- Designed the dashboard so every user sees results scored against their own skills, seniority preference, and keyword rules instead of a generic feed, giving users a fully personalized experience from the same dataset.
- Integrated Google's Gemini API to power AI-assisted job filtering and on-demand application strategy suggestions, including a five-model fallback chain so the feature stays reliable when a model hits a rate limit or error.
- Engineered a caching layer so returning users get instant dashboard loads, while data automatically recomputes in the background whenever the underlying job pool or a user's settings change, balancing perceived performance against compute cost.
- Implemented secure, role-based access end to end with Google OAuth and Supabase, including route-level middleware protection and an admin console, with safeguards that prevent privilege escalation through the API.

Echo FM — Autonomous AI Satirical Radio Station

GitHub – Live – Mirror | *Python · GitHub Actions · Supabase · Webhooks · LLM · FFmpeg · Serverless Architecture*

- Designed and shipped a fully autonomous, serverless media pipeline: live news ingestion → LLM script generation → TTS synthesis → LUFS-mastered audio → automated YouTube publishing, running on a 6-hour CRON schedule with zero manual intervention.
- Integrated multiple LLM APIs (Gemini, Groq/Llama-4) with a custom Smart Router and quality validator; built a 3-tier TTS cascade enforcing vocal consistency across 13-segment broadcasts via a Unified Master Engine pattern.

TECHNOLOGIES

Proficient: React.js · TypeScript · JavaScript · Redux/Redux Toolkit · React Query/TanStack Query · Material-UI · Vite · HTML · CSS · SASS · Bootstrap · Git

Familiar: Next.js · Node.js · Express · MongoDB · GraphQL · WebSocket/STOMP · MobX · Sentry · Jira · Jest · Vitest · React Testing Library · AWS · Docker · Webpack

CERTIFICATES

Udacity — React Nanodegree *Jul 2022 – Aug 2022*

freeCodeCamp — JavaScript Algorithms and Data Structures *Jun 2022 – Jul 2022*

LANGUAGES

Arabic — Native | English — Fluent (C1) | German — Beginner (A0)